

Supporting Information (Online Resource 1)

Glioblastoma drugs and a $\text{Ti}_3\text{C}_2\text{O}_2$ MXene carrier: docking controls, tight-binding tests and periodic DFT adsorption

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Table S1 Compounds (structures from PubChem, identity checked by InChIKey) with their Vina score in EGFR (PDB 1M17, kcal mol^{-1}) and ligand efficiency (kcal mol^{-1} per heavy atom). Bortezomib was not docked: AutoDock Vina has no boron atom type.

Compound	Family	PubChem CID	Formula	Vina	LE
Abemaciclib	Other kinase inhibitors	46220502	$\text{C}_{27}\text{H}_{32}\text{F}_2\text{N}_8$	-9.65	0.261
Afatinib	EGFR inhibitors	10184653	$\text{C}_{24}\text{H}_{25}\text{ClFN}_3\text{O}_3$	-8.03	0.236
Axitinib	Other kinase inhibitors	6450551	$\text{C}_{22}\text{H}_{18}\text{N}_4\text{OS}$	-8.56	0.306
Bortezomib	Other targeted agents	387447	$\text{C}_{19}\text{H}_{25}\text{BN}_4\text{O}_4$	–	–
Brigatinib	Other kinase inhibitors	68165256	$\text{C}_{29}\text{H}_{39}\text{ClN}_7\text{O}_2\text{P}$	-8.64	0.216
Buparlisib	Other kinase inhibitors	16654980	$\text{C}_{18}\text{H}_{21}\text{F}_3\text{N}_6\text{O}_2$	-8.34	0.287
Cabozantinib	Other kinase inhibitors	25102847	$\text{C}_{28}\text{H}_{24}\text{FN}_3\text{O}_5$	-8.66	0.234
Carmustine	Alkylating agents	2578	$\text{C}_5\text{H}_9\text{Cl}_2\text{N}_5\text{O}_2$	-4.75	0.396
Cediranib	Other kinase inhibitors	9933475	$\text{C}_{25}\text{H}_{27}\text{FN}_4\text{O}_3$	-8.12	0.246
Cobimetinib	Other kinase inhibitors	16222096	$\text{C}_{21}\text{H}_{21}\text{F}_3\text{IN}_3\text{O}_2$	-9.01	0.300
Dabrafenib	Other kinase inhibitors	44462760	$\text{C}_{23}\text{H}_{20}\text{F}_3\text{N}_5\text{O}_2\text{S}_2$	-8.71	0.249
Dacomitinib	EGFR inhibitors	11511120	$\text{C}_{24}\text{H}_{25}\text{ClFN}_3\text{O}_2$	-8.78	0.266
Erlotinib	EGFR inhibitors	176870	$\text{C}_{22}\text{H}_{23}\text{N}_3\text{O}_4$	-7.27	0.251
Everolimus	Other targeted agents	6442177	$\text{C}_{53}\text{H}_{83}\text{NO}_{14}$	-7.84	0.115
Gefitinib	EGFR inhibitors	123631	$\text{C}_{22}\text{H}_{24}\text{ClFN}_4\text{O}_3$	-8.10	0.261
Larotrectinib	Other kinase inhibitors	46188928	$\text{C}_{21}\text{H}_{22}\text{F}_2\text{N}_6\text{O}_2$	-9.19	0.296
Lenvatinib	Other kinase inhibitors	9823820	$\text{C}_{21}\text{H}_{19}\text{ClN}_4\text{O}_4$	-8.10	0.270
Lomustine	Alkylating agents	3950	$\text{C}_9\text{H}_{16}\text{ClN}_3\text{O}_2$	-6.25	0.417
Marizomib	Other targeted agents	11347535	$\text{C}_{15}\text{H}_{20}\text{ClNO}_4$	-7.00	0.333
Nimustine	Alkylating agents	39214	$\text{C}_9\text{H}_{13}\text{ClN}_6\text{O}_2$	-6.05	0.336
Osimertinib	EGFR inhibitors	71496458	$\text{C}_{28}\text{H}_{33}\text{N}_7\text{O}_2$	-8.14	0.220
Palbociclib	Other kinase inhibitors	5330286	$\text{C}_{24}\text{H}_{29}\text{N}_7\text{O}_2$	-9.11	0.276
Paxalisib	Other kinase inhibitors	57384863	$\text{C}_{18}\text{H}_{22}\text{N}_8\text{O}_2$	-8.10	0.289
Pazopanib	Other kinase inhibitors	10113978	$\text{C}_{21}\text{H}_{23}\text{N}_7\text{O}_2\text{S}$	-8.53	0.275
Procabazine	Alkylating agents	4915	$\text{C}_{12}\text{H}_{19}\text{N}_3\text{O}$	-6.47	0.404
Regorafenib	Other kinase inhibitors	11167602	$\text{C}_{21}\text{H}_{15}\text{ClF}_4\text{N}_4\text{O}_3$	-8.77	0.266
Ribociclib	Other kinase inhibitors	44631912	$\text{C}_{23}\text{H}_{30}\text{N}_8\text{O}$	-8.45	0.264
Selumetinib	Other kinase inhibitors	10127622	$\text{C}_{17}\text{H}_{15}\text{BrClFN}_4\text{O}_3$	-8.11	0.300
Sorafenib	Other kinase inhibitors	216239	$\text{C}_{21}\text{H}_{16}\text{ClF}_3\text{N}_4\text{O}_3$	-8.52	0.266
Sunitinib	Other kinase inhibitors	5329102	$\text{C}_{22}\text{H}_{27}\text{FN}_4\text{O}_2$	-7.86	0.271
Temozolomide	Alkylating agents	5394	$\text{C}_6\text{H}_6\text{N}_6\text{O}_2$	-5.98	0.427
Trametinib	Other kinase inhibitors	11707110	$\text{C}_{26}\text{H}_{23}\text{FIN}_5\text{O}_4$	-8.55	0.231
Vorinostat	Other targeted agents	5311	$\text{C}_{14}\text{H}_{20}\text{N}_2\text{O}_3$	-6.62	0.349

Table S2 xtb calculations attempted on finite $\text{Ti}_3\text{C}_2\text{O}_2$ flakes. uhf, number of unpaired electrons; T_{el} , electronic temperature (K).

Structure	R un	Meth od	Charge	uhf	T_{el}	Options	SCF converged
Ti55C24O20 flake or smaller cut	opt	GFN 2	0	0	150 0		no
Ti55C24O20 flake or smaller cut	sp	GFN	0	0	300	--acc 0.1	no

		2		0		
Ti55C24O20 flake or smaller cut	sp	GFN 1	0	0	150 0	no
earlier non-stoichiometric Ti12C7O14 cluster (discarded)	sp	GFN 2	0	0	300	yes
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	150 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	1	150 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	150 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	300	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	150 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	300	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	-2	0	100 0	yes
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	2	100 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	4	100 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	6	100 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	2	0	100 0	no
Ti55C24O20 flake or smaller cut	sp	GFN 2	4	0	100 0	yes
Ti55C24O20 flake or smaller cut	sp	GFN 2	-2	0	100 0	yes
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	100 0	restarted from dianion charges yes
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	300 0	damped no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	300 0	damped gasteiger guess no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	2	300 0	damped no
Ti55C24O20 flake or smaller cut	sp	GFN 1	0	0	300 0	damped no
Ti55C24O20 flake or smaller cut	sp	GFN 2	0	0	150 0	no

14

15 [Table S3 (DFT convergence checks) is generated when

16 results/quantum/dft_convergence_checks.csv exists.]

17 Table S4 Every output mode of the two erlotinib redocking controls in PDB 1M17: Vina score and heavy-atom
18 RMSD to the crystal pose (symmetry-aware, no re-alignment).

Control	Mode	Vina (kcal mol ⁻¹)	RMSD (Å)
self-redock, crystal conformation	1	-7.020	5.94
self-redock, crystal conformation	2	-6.935	7.96
self-redock, crystal conformation	3	-6.885	9.45
self-redock, crystal conformation	4	-6.836	6.03
self-redock, crystal conformation	5	-6.829	4.75
self-redock, crystal conformation	6	-6.808	1.13
self-redock, crystal conformation	7	-6.785	8.81
self-redock, crystal conformation	8	-6.758	2.33
self-redock, crystal conformation	9	-6.745	7.42
production protocol, from SMILES	1	-7.284	7.61
production protocol, from SMILES	2	-7.214	4.92
production protocol, from SMILES	3	-7.176	5.97
production protocol, from SMILES	4	-7.163	6.02
production protocol, from SMILES	5	-7.130	8.44
production protocol, from SMILES	6	-7.042	8.32
production protocol, from SMILES	7	-7.013	1.44
production protocol, from SMILES	8	-6.882	7.99

production protocol, from SMILES	9	-6.833	7.25
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Table S5 Energy of the periodic 4×4 Ti₃C₂O₂ slab per formula unit against the in-plane lattice constant, GFN2-xTB and GFN1-xTB (tblite, Γ point, 1500 K). The GFN1-xTB values at 2.90 and 2.95 Å (−3082 to −3039 eV per f.u.) are unphysical, about 6.4 times the energy at the other lattice constants. PBE-D3 reference (last row): variable-cell relaxation of the primitive 1×1 cell with the settings of the adsorption calculations, 12×12×1 k-points, residual in-plane pressure −0.15 kbar; the slab uses $a = 3.03$ Å, 0.3% larger. A tetrahedron-method density of states on a 24×24×1 grid gives 1.0 states eV^{−1} per formula unit at the Fermi level: the material is metallic.

Method	a (Å)	E (eV per f.u.)	Status
GFN2-xTB	2.60	−463.453	ok
GFN2-xTB	2.65	−464.916	ok
GFN2-xTB	2.70	−495.901	ok
GFN2-xTB	2.75	−493.797	ok
GFN2-xTB	2.80	−491.783	ok
GFN2-xTB	2.85	−489.844	ok
GFN2-xTB	2.90	−487.972	ok
GFN2-xTB	2.95	−486.160	ok
GFN2-xTB	3.00	−484.408	ok
GFN2-xTB	3.05	−482.714	ok
GFN2-xTB	3.10	−481.078	ok
GFN2-xTB	3.15	−479.499	ok
GFN2-xTB	3.20	−477.976	ok
GFN2-xTB	3.25	−476.510	ok
GFN1-xTB	2.60	—	SCF not converged in 500 cycles
GFN1-xTB	2.65	—	SCF not converged in 500 cycles
GFN1-xTB	2.70	—	SCF not converged in 500 cycles
GFN1-xTB	2.75	—	SCF not converged in 500 cycles
GFN1-xTB	2.80	—	SCF not converged in 500 cycles
GFN1-xTB	2.85	—	SCF not converged in 500 cycles
GFN1-xTB	2.90	−3039.118	ok
GFN1-xTB	2.95	−3081.802	ok
GFN1-xTB	3.00	−479.220	ok
GFN1-xTB	3.05	−479.069	ok
GFN1-xTB	3.10	−478.815	ok
GFN1-xTB	3.15	−478.472	ok
GFN1-xTB	3.20	−478.056	ok
GFN1-xTB	3.25	−477.855	ok
PBE-D3	3.022	—	reference optimum

[Table S6 (per-drug DFT details) is generated when the DFT campaign is complete.]